Luca Walz

Software Developer @ IBM | watsonx.data intelligence

GitHub github.com/lucawalz · LinkedIn linkedin.com/in/luca-walz · lucawalz.dev · EU citizen

Software Developer at IBM on watsonx.data intelligence, working on backend services in Java. Outside of work: Go, Kubernetes, NixOS and a homelab run entirely from Git.

EXPERIENCE

IBM · Böblingen, Hybrid

Software Developer · Full-time — Since Sep 2026
Working Student · Part-time — Sep 2025 - Dec 2025
Software Development Intern · Full-time — Mar 2025 - Aug 2025

As intern and working student: built API test suites and backend testing workflows for IBM Knowledge Catalog, validating enterprise-scale microservices alongside the dev and QA teams.

Java · Jenkins · Microservices · API testing

EDUCATION

Hochschule der Medien Stuttgart · B.Sc. Media Informatics (Medieninformatik) — 2023 - 2026

Bachelor thesis on autonomous cluster remediation with LLM agents.

Freie Waldorfschule, Böblingen · Abitur — 2020 - 2022

Burnaby Mountain Secondary School, Vancouver · Exchange year, Canada — 2019 - 2020

SELECTED PROJECTS

vigil — Python, Go

My bachelor thesis: a multi-agent system that diagnoses and repairs faults in a Kubernetes cluster on its own. The LLM agents act through typed MCP tools and are reversible by construction, with Flux GitOps and NixOS generations giving every repair a clean way back.

bedrock — Nix

A bare-metal Kubernetes homelab that lives entirely in Git. Three NixOS nodes run K3s, and Flux reconciles the cluster from the repository instead of by hand.

horizon — Go

The active half of the homelab: a Kubernetes operator that leases temporary Hetzner Cloud capacity. A lease says how much, from where and for how long, and the controller guarantees the machines are gone at the deadline even if the cluster and the network are not there to help.

sentio-systems · Team Lead and Lead Developer — Java, TypeScript, Python

A wildlife and weather monitoring platform. Raspberry Pi field devices feed sensor and camera data to a Spring Boot backend, where AI classifies the animals down to the bird species, with a React dashboard and an agentic chatbot to explore it all.

SKILLS

Languages	Go, Python, Java, TypeScript
Runtimes and frameworks	Spring, React, Nix
Infrastructure and data	Linux, Kubernetes, Docker, Terraform, Postgres, Redis, Prometheus, Grafana, Raspberry Pi

VOLUNTEERING

Student Mentor · Hochschule der Medien, Stuttgart — Mar 2024 - Aug 2024

Supporting first-semester students in finding their footing on campus and in their studies.

Downhill Trainer — Since 2014

Scout — Since 2008

Long-term member, organising outdoor excursions with larger groups.

LANGUAGES

German (Native) · English (C2) · Russian (A1)